Spring 2026 *Name*: _____ ID: _____
CSCI 4311/5311 Homework # 1

Due: **March 1, 2026 (11:59 pm), via Canvas.**

The rules:

- All work must be your own. You are not allowed to work in teams on this assignment, nor are you permitted to use materials from previous offerings of this course.
- **Format:** Submit a single file (via Canvas) containing a PDF file. Email me (ayn@cs.uno.edu) the assignment only if Canvas is not working.
- You may use the textbook and lecture notes, but do NOT search the Internet for solutions.
- The submission deadline is strict, so please ensure you submit on time.

Total Marks = 100

(Q1) [5x3 = 15 points]

Suppose Host A wants to send a large file to Host B. The path from Host A to Host B has three links of rates R1 = 500 kbps, R2 = 2 Mbps, and R3 = 1 Mbps.

a-) Assuming no other traffic in the network, what is the throughput for the file transfer?

b-) Suppose the file is 4 million bytes. Dividing the file size by the throughput, roughly how long will it take to transfer the file to Host B? (Ignore all other delays, just focus on transmission delay)

c-) Repeat (a) and (b), but now with R2 reduced to 100 kbps.

(Q2) [10 points]

Why will two ISPs at the same level of the hierarchy often prefer peering agreement with each other? How does an IXP earn money?

(Q3) [15 points]

Suppose you would like to urgently deliver 40 terabytes data from Boston to Los Angeles. You have available a 100 Mbps dedicated link for data transfer. Would you prefer to transmit the data via this link or instead use FedEx overnight delivery? Explain.

Assumptions:

- There is a direct fiber link between Boston and Los Angeles offices.
- Ignore processing, queueing, and propagation delay. Only calculate transmission delay.
- If you choose FedEx, you will give the packet at 9 am, and it will be delivered in the next day at 9 am.

(Q4) [4x5 = 20 points]

Consider the following string of ASCII characters that were captured by Wireshark when the browser sent an HTTP GET message (i.e., this is the actual content of an HTTP GET message). The characters <cr><lf> are carriage return and line-feed characters (that is, the italized character string <cr> in the text below represents the single carriage-return character that was contained at that point in the HTTP header). Answer the following questions, indicating where in the HTTP GET message below you find the answer.

```
GET /cs453/index.html HTTP/1.1<cr><lf>Host: gai
a.cs.umass.edu<cr><lf>User-Agent: Mozilla/5.0 (
Windows;U; Windows NT 5.1; en-US; rv:1.7.2) Gec
ko/20040804 Netscape/7.2 (ax) <cr><lf>Accept:ex
t/xml, application/xml, application/xhtml+xml, text
/html;q=0.9, text/plain;q=0.8, image/png,*/*;q=0.5
<cr><lf>Accept-Language: en-us, en;q=0.5<cr><lf>Accept-
Encoding: zip, deflate<cr><lf>Accept-Charset: ISO
-8859-1, utf-8;q=0.7,*;q=0.7<cr><lf>Keep-Alive: 300<cr>
<lf>Connection:keep-alive<cr><lf><cr><lf>
```

a. What is the URL of the document requested by the browser?

b. What version of HTTP is the browser running?

c. Does the browser request a non-persistent or a persistent connection?

d. What is the IP address of the host on which the browser is running? (Trick question)

e. What type of browser initiates this message? Why is the browser type needed in an HTTP request message?

(Q5) [5x4 = 20 points]

Hosts A and B are communicating over a TCP connection, and Host B has already received from A all bytes up through byte 500. Suppose Host A then sends three segments to Host B back-to-back. The first, second and third segments contain 100, 70, and 80 bytes of data respectively. In the first segment, the sequence number is 500, the source port number is 1000, and the destination port number is 2000. Host B sends an acknowledgment whenever it receives a segment from Host A.

a-) In the third segment sent from Host A to B, what are the sequence number, source port number, and destination port number? Briefly explain.

b-) In Host B receives all three segments successfully, what are the acknowledgment number, source port number, and destination port number in the last segment that Host B sent? Briefly explain.

c-) If Host B receives the segments in the following order: “2-1-3”. What is the first ACK number that Host B sends after receiving segment number 2? Briefly explain.

d-) Assume that Host B received all 3 segments without any problem and sent ACK segments for each arriving segment. Assume that the ACK segment 2 is lost. What will Host A do and why? E.g., retransmit packets, retransmit a packet or, do not retransmit the packet.

(Q6) [20 points]

Assume that we have a GBN protocol with a window size 3. We are trying to send 8 packets total (e.g. [0,1,2,3,4,5,6,7]).
Show both sender and receiver side for given cases. For the receiver side, write what is the ACK number. In the sender side, show what is the packet number. Also, show the sender window each time it changes. Finally, show what happens after the timeout, e.g. Sender retransmits some packets, receiver sends ACK, show the values. Explain your steps briefly.

Main assumptions for the question:

- Assume that the timeout does not occur prematurely.
- We do not have network congestion or additional delay.
- You can reference slide #49.

a-)[6 points] Assume that all packets including ACK packets delivered without error. Assume that the sender sent all packets before timeout.

b-)[7 points] Assume that only packet #4 loss during the first transmission. All other packets and retransmission etc. delivered correctly.

c-)[7 points] Assume that only ACK packet #3 loss during the first transmission. All other packets and retransmission etc. delivered correctly.